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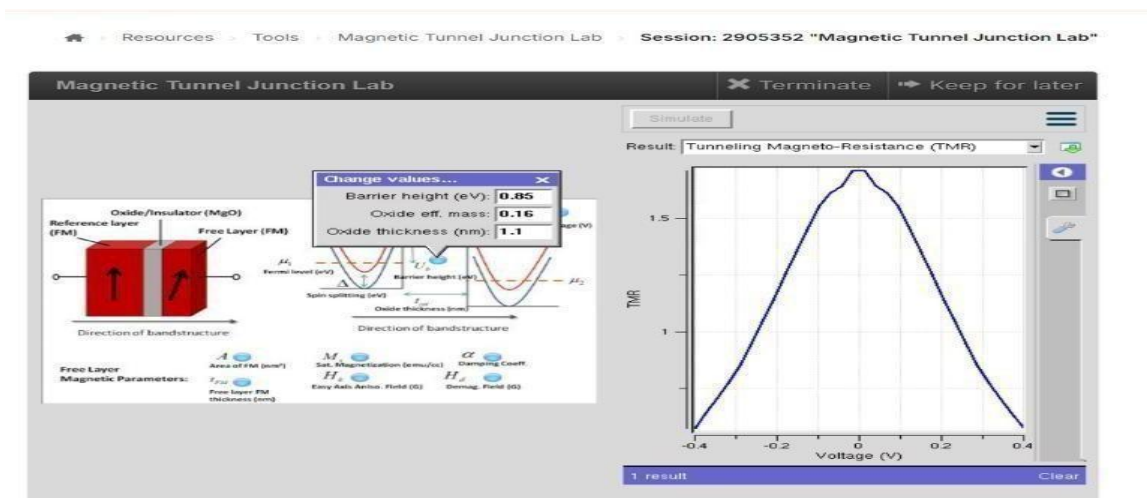
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Exp. 8. Simulation of Tunneling Magnetoresistance (TMR) by Varying Barrier Height using Nano HUB

Aim:

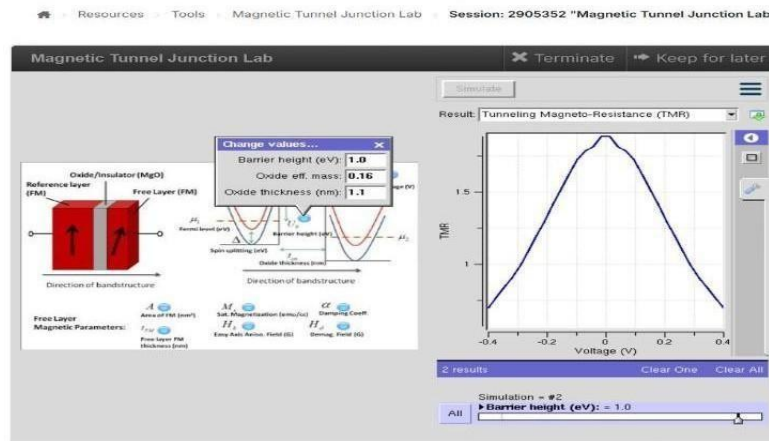
To obtain the Tunnelling Magnetoresistance (TMR) by varying barrier height.

Case Study 1: Characteristics of TMR with barrier height of 0.85eV



Output: The graph illustrates the variation of Tunneling Magnetoresistance (TMR) with applied voltage for a barrier height of 0.85 eV. The TMR reaches its maximum value at 0 V and decreases gradually and symmetrically as the applied voltage increases in both the positive and negative directions. This behavior indicates that electron spin polarization is highest under zero-bias conditions, resulting in maximum tunneling efficiency. As the bias voltage increases, the tunneling probability decreases, leading to a reduction in the TMR ratio. This characteristic demonstrates the influence of barrier height on the spin-dependent transport properties of magnetic tunnel junctions.

Case Study 2: Characteristics of TMR with barrier height of 1.0eV



Output: The graph shows the Tunneling Magnetoresistance (TMR) as a function of applied voltage for a barrier height of 1.0 eV. Similar to the previous case, the TMR exhibits a peak near 0 V and decreases symmetrically with increasing positive and negative bias voltage. The higher barrier height provides stronger confinement for tunneling electrons, influencing the electron transport characteristics and maintaining stable spin-dependent conduction at low bias voltages. This result confirms that the barrier height plays a significant role in determining the TMR performance of magnetic tunnel junction devices.

Comparison and Conclusion

The TMR characteristics for 0.85 eV and 1.0 eV barrier heights were successfully simulated and compared. In both cases, the TMR reaches its maximum value at zero bias and decreases symmetrically as the applied voltage increases. However, the 1.0 eV barrier height provides better electron confinement and more stable tunneling characteristics, making it more suitable for high-performance spintronic devices. Overall, the simulation

demonstrates that barrier height significantly influences the tunneling magnetoresistance and spin-dependent electron transport, which are essential for the design and optimization of magnetic tunnel junctions (MTJs), MRAM, and other spintronic applications.